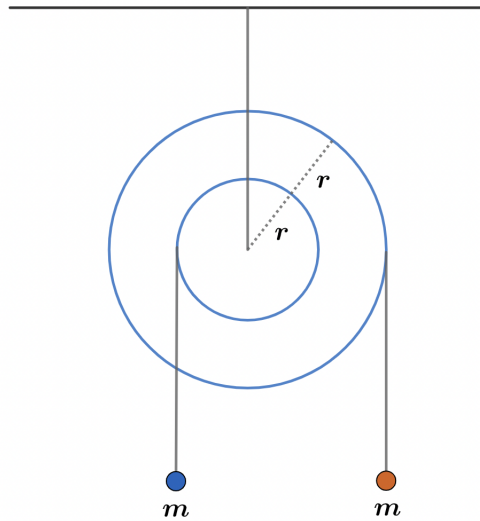


2022A F=ma Exam: Problem 12

Kevin S. Huang



By Newton's 2nd law,

$$F_1 = ma_1$$

$$F_2 = ma_2$$

The angular acceleration of the pulley is related to the tangential acceleration by

$$\alpha = \frac{a_1}{r} = \frac{a_2}{2r}$$

Thus,

$$\frac{F_2}{F_1} = \frac{a_2}{a_1} = 2$$

so the answer is D.